

The Nobel Prize in Physiology or Medicine is given for discoveries deemed particularly important in the fields of physiology or medicine. In 1906, this award was given to two biologists for their groundbreaking work in the field of neuroanatomy. Despite using the same technique to visualize neurons, the two biologists formed opposing theories regarding organization of the nervous system.

Hypothesis 1: Reticular Theory

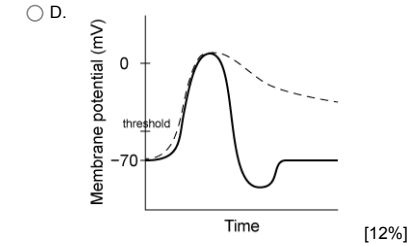
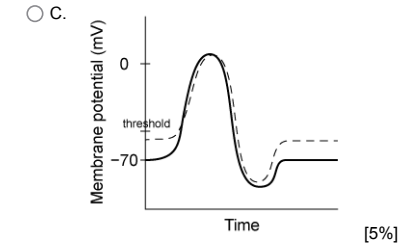
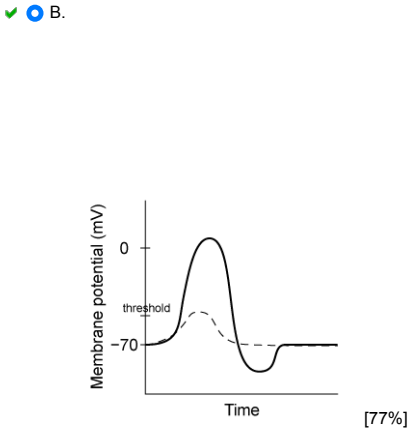
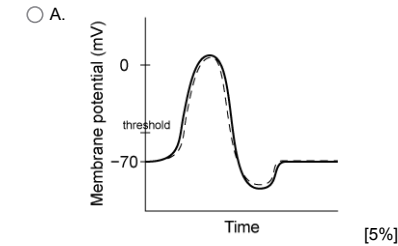
Biologist 1 developed a neural staining technique that was carried out by fixing brain tissue from older animals with primarily myelinated nerve fibers in potassium dichromate ($K_2Cr_2O_7$) before treating this tissue with silver nitrate ($AgNO_3$). A silver chromate (Ag_2CrO_4) precipitate deposited within the neurons, making them appear black. This silver impregnation method was revolutionary because of two features. First, only a random subset of neurons within a given tissue sample (~1% of the total cell population) was subject to silver impregnation. Second, the impregnated cells were stained in their entirety, allowing for visualization of the complete neuron. Biologist 1 concluded that his findings supported the idea that the nervous system is a single reticulum of fused neural processes that transmit information via cytoplasmic continuity.

Hypothesis 2: Neuron Doctrine

Biologist 2 modified the silver impregnation method by first subjecting the brain tissue to two rounds of staining ("double impregnation") rather than one. Where Biologist 1 studied the brain tissue of older animals with myelinated nerve fibers, Biologist 2 employed the silver stain on brain tissue from younger animals, which made it easier for the silver to impregnate the neural processes due to a relative lack of myelination. Biologist 2 reasoned that the nervous system is not a single web of interconnected fibers but rather is made up of discrete neural units that communicate via close, but not continuous, contact.

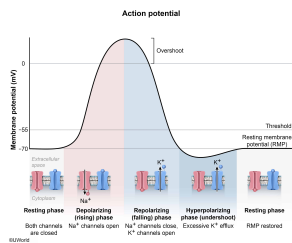
The neuron doctrine was later proven correct with the advent of electron microscopy, which allowed scientists to visualize the synapse (ie, the point of communication between two distinct neurons). Further investigation allowed researchers to ultimately discover that there are two types of synapses: electrical and chemical.

A neuron is exposed to a chemical that prevents the passage of Na^+ ions through voltage-gated Na^+ channels. Which of the following graphs best represents the local membrane potential change of this neuron upon injection of a positive current that continues until the membrane reaches the threshold value? (Note: The solid line represents the action potential after normal stimulation, and the dashed line represents the action potential after exposure to the chemical.)



Correct
77% answered correctly

Explanation:



An action potential (AP) is a rapid electrical impulse that travels from the neuron's cell body to its axon terminal. It consists of a series of membrane potential changes, as follows:

- Resting membrane potential (RMP):** The membrane potential of the resting neuron is -70 mV (ie, the inside of the neuron is 70 mV more negative than the extracellular space). RMP is maintained by the [Na⁺/K⁺ pump](#) and K⁺ "leak" channels, which always allow passive diffusion of K⁺ ions across the membrane. Ion channels that open or close based on changes in membrane potential (voltage-gated channels) are closed at RMP.
- Threshold:** An excitatory stimulus causes several nearby voltage-gated Na⁺ channels to open, allowing Na⁺ ions to rush into the cell. Consequently, the membrane potential becomes more positive than the RMP (ie, depolarized). If the neuron depolarizes to its specific *threshold* value, an AP is fired. If the threshold is not reached, no AP is fired and the RMP is restored.
- Rising phase:** At threshold, the remaining voltage-gated Na⁺ channels open, resulting in rapid depolarization of the local membrane. This rise in membrane potential triggers a positive feedback loop, opening more Na⁺ channels in adjacent segments to propagate the AP down the axon.
- Overshoot:** The ongoing flood of Na⁺ into the cell causes the AP to reach its peak (overshoot), where the membrane potential is most positive.
- Falling phase:** Voltage-gated Na⁺ channels close, and voltage-gated K⁺ channels open fully. K⁺ then rushes out of the cell, causing membrane repolarization and restoring RMP.
- Undershoot:** Excessive K⁺ efflux causes the membrane potential to fall below the RMP (hyperpolarize), and the local membrane enters a refractory period ([absolute or relative](#)).
- RMP restoration:** Both voltage-gated Na⁺ and K⁺ channels are inactive. The membrane potential returns to -70 mV and is maintained by the Na⁺/K⁺ pump until another action potential is initiated.

In this scenario, the chemical **prevents** voltage-gated **Na⁺ channels from opening** regardless of membrane depolarization. Therefore, even when the threshold is reached, the channels *cannot* open, the rising phase is [not effectively initiated](#), and no AP is generated. After stimulation, the membrane potential simply returns to RMP, as in the graph shown in Choice B. [The other graphs](#) do *not* display these characteristics.

Educational objective:

An action potential is fired in an all-or-none manner based on the cell's membrane potential. If the threshold is reached, voltage-gated ion channels open and the membrane successively depolarizes and repolarizes. The resting membrane potential is restored by the Na⁺/K⁺ pump.

Time spent: 0 sec
Subject:
Biology

QID: 400954
System:
Endocrine and Nervous Systems